

HEKKINGEN FYR, Norway

WMO: 010150

Lat: 69.601N

Lon: 17.831E

Elev: 46

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	15.7	18.0	5.1	7.1	17.5	7.9	8.2	19.7	50.8	40.0	45.7	34.9	22.5	160	0.853

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.1	66.1	57.9	63.4	56.2	60.8	54.9	58.8	64.4	57.0	62.1	55.4	60.0	13.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
55.9	66.6	61.7	54.1	62.4	59.8	52.7	59.2	58.2	25.6	64.7	24.5	62.4	23.5	60.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 42.2	34.8	30.2	12.7	71.9	4.5	4.5	9.4	75.1	6.8	77.7	4.2	80.2	1.0	83.5		
(5)			11.0	61.6	4.3	2.6	7.9	63.5	5.4	65.0	3.0	66.5	-0.1	68.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.7	29.9	29.9	31.5	37.2	43.7	48.3	53.8	53.6	49.0	41.7	36.1	33.3		
	DBStd	10.18	6.69	6.74	5.62	5.18	5.36	4.87	4.75	4.27	4.95	5.56	5.60	6.08		
	HDD50	3737	624	564	573	384	207	89	12	12	77	261	417	517		
	HDD65	8862	1089	984	1038	834	660	502	350	355	480	721	867	982		
	CDD50	352	0	0	0	0	13	37	129	122	46	5	0	0		
	CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH74	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	14.8	19.8	19.1	17.0	13.6	12.3	10.4	9.4	10.3	13.7	15.3	17.5	19.1		
	PrecAvg	41.1	3.7	3.3	3.6	2.5	2.0	2.2	2.7	3.0	4.3	5.3	4.1	4.1		
(15) Precipitation	PrecMax	50.6	9.0	6.4	8.5	6.3	5.5	4.4	5.4	7.4	8.9	8.3	9.0	7.2		
	PrecMin	31.6	0.0	0.6	0.8	0.0	0.2	0.5	0.6	1.1	1.1	0.9	0.4	0.0		
	PrecStd	4.7	2.5	1.3	2.2	1.3	1.2	0.9	1.4	1.5	2.2	2.0	2.2	2.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.7	46.8	47.3	53.2	62.7	68.4	70.9	67.9	63.0	57.1	50.7	48.8		
		MCWB	41.6	41.8	40.1	44.6	51.4	55.4	60.5	59.7	54.1	49.7	46.1	43.4		
	2%	DB	43.8	44.1	44.6	49.4	57.6	62.1	66.8	64.8	59.7	53.7	48.1	46.4		
		MCWB	39.4	39.2	39.0	42.9	49.3	53.7	58.6	57.9	53.1	48.1	44.3	41.9		
	5%	DB	41.5	42.1	42.1	46.6	54.1	58.5	64.1	62.2	57.7	51.5	45.9	43.8		
		MCWB	37.7	37.9	37.2	41.3	46.8	52.1	56.9	56.3	52.0	46.8	41.5	39.4		
	10%	DB	38.9	39.2	39.5	44.4	51.2	55.5	61.3	59.9	55.7	49.4	43.8	41.4		
		MCWB	35.5	35.8	35.6	39.8	45.2	50.5	55.5	55.1	50.9	45.2	40.0	37.3		
	0.4%	WB	42.3	42.7	41.4	45.3	52.8	57.0	61.5	61.8	55.9	50.3	46.8	44.3		
		MCDB	45.6	46.0	45.6	51.6	61.0	66.4	68.7	65.4	61.0	55.0	49.7	48.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	39.9	39.6	39.4	43.3	49.5	54.4	59.3	58.9	54.1	48.8	44.8	42.1		
		MCDB	43.3	43.3	43.6	48.2	56.7	61.0	65.7	63.2	58.3	53.0	48.0	45.9		
	5%	WB	37.8	38.0	37.7	41.7	47.5	52.6	57.5	57.0	52.8	47.1	42.0	39.7		
		MCDB	41.2	41.6	41.6	46.1	53.3	57.9	63.4	61.1	56.7	50.9	45.3	43.4		
	10%	WB	35.6	35.9	35.8	40.1	45.6	50.8	55.8	55.5	51.4	45.6	40.2	37.5		
		MCDB	38.8	39.2	39.3	44.1	50.4	55.0	60.5	59.5	55.1	49.1	43.6	41.0		
	5% DB	MDBR	5.8	5.8	5.7	6.9	7.5	6.4	7.1	6.6	5.8	5.0	5.2	5.8		
		MCDBR	8.3	8.8	8.0	9.4	11.8	11.4	12.5	10.4	8.5	7.5	7.6	8.8		
		MCWBR	7.0	7.1	6.4	6.1	6.7	6.7	7.1	6.2	5.1	5.6	6.1	7.2		
		MCWBR	8.2	8.8	7.6	9.0	11.2	10.3	11.4	9.3	7.8	7.2	7.2	8.6		
(35) Clear-Sky Solar Irradiance	taub		0.374	0.273	0.318	0.334	0.343	0.342	0.346	0.335	0.319	0.278	N/A	N/A		
	taud		2.586	2.101	2.193	2.320	2.438	2.496	2.516	2.527	2.528	2.219	N/A	N/A		
	Ebn at Noon		14	142	215	254	268	272	266	255	226	140	0	0		
	Edh at Noon		3	13	24	29	29	28	27	23	18	11	0	0		
(44) All-Sky Solar Radiation	RadAvg		3	81	399	958	1438	1553	1390	983	518	158	0	0		
	RadStd		0	16	57	98	102	203	138	81	55	25	0	0		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page